

GRPO

What is Reference Tuning?

What is RL?

What is a PPO?

What is RLHF?

What is GRPO?

GRPO (Group Relative Policy Optimization) is a reinforcement learning method used in LLM alignment that optimizes model behavior based on relative comparisons within a group of responses, instead of relying on absolute reward scores.

GRPO learns by comparing multiple responses and improving based on relative ranking instead of absolute reward values.

Key Idea

- PPO → optimizes using absolute reward values
- GRPO → optimizes using relative ranking within a group

Instead of asking:

"How good is this answer?"

GRPO asks:

"Which answer is best among this group?"

Then GRPO (2024–2025, used by DeepSeek)

👉 Evolution of PPO + DPO ideas

Group-based comparison + RL optimization

👉 Key innovation

No reward model + multiple samples per prompt

Why GRPO?

- More stable than PPO
- Less dependent on precise reward modeling
- Better suited for LLM alignment scenarios
- Works well when:
 - evaluation rule exists
 - multiple samples possible

Positioning

- PPO → traditional RL method (still used, especially in RLHF)
- GRPO → newer alternative focusing on relative comparison
- DPO / ORPO → often preferred today (no RL loop, simpler training)

GRPO is a promising alternative to PPO in certain alignment setups

How GRPO Works

1. Take a prompt
2. Generate multiple responses (e.g., 4–8)
3. Compare or rank the responses
4. Compute relative advantage within the group
5. Update the model based on relative performance

Core Intuition

- PPO → “Assign a score to each answer”
- GRPO → “Compare answers and learn which is better”

How GRPO Works (Step-by-Step)

Step 1: Given a prompt

User: Explain AI

Step 2: Generate multiple responses

A: AI is simulation of intelligence

B: AI is random stuff

C: AI is machines learning patterns

Step 3: Compare within group

Instead of reward model:

Rank them:

$C > A > B$

Step 4: Compute relative advantage

GRPO computes:

“How much better is A compared to others?”

Based on group ranking

Step 5: Update policy

Increase probability of better responses

Decrease probability of worse ones

Similar to PPO

But no reward model

Intuition (Super Simple)

PPO:

"This answer = score 8/10"

GRPO:

"This answer is better than others"

Real Example

Prompt:

Write a polite email

Generated:

- A → rude
- B → okay
- C → very polite

GRPO learns:

$C > B > A$

Update:

- Increase C
- Decrease A

Why DeepSeek Used GRPO

Problem DeepSeek faced

They were training huge LLMs (DeepSeek-R1)

PPO issues:

- Reward model training cost
- Instability
- Slow

Problem with PPO (RLHF)

- Needs Reward Model
- Expensive training
- Complex pipeline

DeepSeek Solution = GRPO

Instead of:

Generate → Reward Model → PPO

They did:

Generate multiple responses → Compare → Update

🔥 CRITICAL FIX (DeepSeek part)

❌ Your statement:

- "No reward model + self learning only"

✅ Correct:

- DeepSeek used rule-based + task-specific rewards
- Not purely "model learning from itself blindly"

👉 Example:

- math problems → correctness check
- reasoning → format / logic check

❌ Your statement:

- "GRPO = no reward model"

✅ Correct version:

- GRPO can work without a trained reward model
- BUT still needs some scoring signal (heuristic / rule / proxy reward)

👉 Matlab:

- Reward model ❌
- Reward signal ✅

MOST IMPORTANT CORRECTION (Conceptual)

❌ Wrong understanding:

- "GRPO = no reward"

✅ Correct:

- GRPO still needs signal to compare outputs
- It just avoids training a separate reward model

Why GRPO is better (for them)

1. No Reward Model

- Removes entire pipeline

2. Self-improving loop

Model generates multiple answers → learns from its own outputs

3. More stable

- Uses relative advantage
- Not absolute noisy reward

4. Cheaper & faster

No reward model training

Less infrastructure

⚠️ WHERE YOU WERE SLIGHTLY WRONG / INCOMPLETE

❌ 1. "No reward model" ko galat samjha

👉 Tu bola:

GRPO = no reward model

👉 Correct:

GRPO = no trained reward model

BUT still needs reward signal (score)

✓ Paper me:

DeepSeek used rule-based reward (correct/incorrect)

❌ 2. Value model removal ka exact reason missing tha

👉 Tu bola:

GRPO simpler hai

👉 Paper actually bolta hai:

Value model remove karke

group mean se advantage calculate hota hai

👉 Ye core innovation hai

❌ 3. Advantage formula intuition missing

👉 Tu bola:

"relative advantage"

👉 Paper me:

Proper normalization hota hai:

👉 $(\text{reward} - \text{mean}) / \text{std}$

✓ Ye hi "group relative" ka actual meaning hai

❌ 4. KL penalty placement nuance missing

👉 Tu bola:

similar to PPO

👉 Correct:

PPO → KL inside advantage

GRPO → KL separate term

👉 Small but important research detail